

Project Design Phase-II
Data Flow Diagram & User Stories

Date	28 January 2026
Team ID	LTVIP2026TMIDS86494
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns Using Tableau
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

DFD – Level 0: Context Diagram



DFD – Level 1: Detailed Diagram



DFD – Level 2: Analysis & Visualization Breakdown



DFD – Level 2: Analysis & Visualization Breakdown



User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Customer (Mobile user)	Data Access	USN-1	As a user, I want to load electricity consumption data into Tableau so that I can analyze it visually.	Data is successfully imported and visible in Tableau	High	Sprint-1
Customer (Mobile user)	Data Preprocessing	USN-2	As a user, I want to clean and preprocess the data to remove missing or inconsistent values.	Cleaned dataset is ready for analysis	High	Sprint-1
Customer (Mobile user)	Time-based Analysis	USN-3	As a user, I want to analyze electricity consumption trends over time (monthly/yearly).	Line charts show correct time-based trends	High	Sprint-1
Customer (Mobile user)	Regional Analysis	USN-4	As a user, I want to compare electricity consumption across different states or regions.	Bar/Map visuals show region-wise consumption	High	Sprint-2
Customer (Mobile user)	Sector-wise Analysis	USN-5	As a user, I want to analyze electricity usage by residential, commercial, and industrial sectors.	Sector-wise comparison is displayed clearly	Medium	Sprint-2
Customer (Mobile user)	Dashboard	USN-6	As a user, I want to view all key insights in a single interactive dashboard.	Dashboard loads with multiple visualizations	High	Sprint-2
Customer (Mobile user)	Filters & Interaction	USN-7	As a user, I want to filter data by year, state, and sector for better analysis.	Filters update visuals dynamically	Medium	Sprint-3
Customer (Mobile user)	Comparative Analysis	USN-8	As a user, I want to compare electricity consumption between different years (2019 vs 2020).	Comparison charts show accurate differences	High	Sprint-3
Customer (Mobile user)	Storytelling	USN-9	As a user, I want to create a Tableau story to explain insights step-by-step.	Story contains ordered visuals with explanations	Medium	Sprint-3
Customer (Mobile user)	Reporting	USN-10	As an evaluator, I want to review dashboards and insights clearly for assessment.	Dashboards and stories are easy to understand	High	Sprint-4